

Modern Hotel Network Design & Implementation

- **Infrastructure & VLAN Architecture:** Designed a 3-floor hotel network using 3 routers (located in the IT server room) interconnected via **Serial DCE cables** (10.10.10.0/30, 10.10.10.4/30, 10.10.10.8/30). Implemented 8 isolated VLANs across floors for each department (Admin, IT, Sales, HR, Finance, Logistics, Store, Reception), with local Wi-Fi access and printers.
- **Routing & Dynamic Services:** Configured **OSPF** for inter-floor routing and full end-to-end connectivity. Set up router-based **DHCP** for dynamic IP addressing across all end devices.
- **Management & Port Security:** Enabled **SSH** across all routers for secure remote access. Implemented **Port Security** on the IT-dept switch (Fa0/1) using mac-address sticky and violation shutdown to restrict access exclusively to Test-PC.

Floor / Location	Department / Role	VLAN Name	VLAN ID	Subnet Network
1st Floor	Logistics	vlan60	60	192.168.6.0/24
1st Floor	Store	vlan70	70	192.168.7.0/24
1st Floor	Reception	vlan80	80	192.168.8.0/24
2nd Floor	Sales	vlan30	30	192.168.3.0/24
2nd Floor	HR	vlan40	40	192.168.4.0/24
2nd Floor	Finance	vlan50	50	192.168.5.0/24
3rd Floor	Admin	vlan20	20	192.168.2.0/24
3rd Floor	IT	vlan10	10	192.168.1.0/24
WAN Connections	Router Link 1 (P2P)	N/A	N/A	10.10.10.0/30
WAN Connections	Router Link 2 (P2P)	N/A	N/A	10.10.10.4/30
WAN Connections	Router Link 3 (P2P)	N/A	N/A	10.10.10.8/30